

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I **JESINTHA[192511410]** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: jesintha[192511410]

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead

introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**.

The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the

compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

ANSWERS:

1. File Transfer using Transport and Data Link Layers

Given:

- File size = 150 MB = 150,000,000 bytes
- Bandwidth = 50 Mbps
- Segment size = 1500 bytes
- Data Link overhead = 40 bytes/frame

Calculation:

- Number of segments = $150,000,000 \div 1500 = \mathbf{100,000 \text{ segments}}$
- Number of frames = **100,000 frames**
- Total overhead = $100,000 \times 40 = \mathbf{4,000,000 \text{ bytes (4 MB)}}$
- Total data transmitted = **154 MB**
- Transmission time = $(154 \times 8) \div 50 = \mathbf{24.64 \text{ seconds}}$

Explanation:

The **Transport Layer** divides the file into segments, ensures reliable delivery through sequencing, acknowledgements, and error recovery. The **Data Link Layer** encapsulates each segment into frames, adds headers and trailers, performs error detection using CRC, and ensures reliable node-to-node communication.

2. Sliding Window Protocol

Given:

- Window size = 6 frames
- Frame size = 1024 bytes
- Total frames = 48

Calculation:

- Number of windows = $48 \div 6 = 8$ **windows**
- One frame is lost in each window.
- Total retransmissions = **8 frames**

Explanation:

Flow control prevents the sender from transmitting data faster than the receiver can process it. This avoids buffer overflow, reduces congestion, and improves communication efficiency.

3. IP Fragmentation

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Calculation:

- Data per fragment = $1500 - 20 = 1480$ bytes
- Number of fragments = $12,000 \div 1480 = 8.11 \approx 9$ **fragments**
- Total header overhead = $9 \times 20 = 180$ bytes

Explanation:

The **Network Layer** fragments packets when they exceed the MTU. Each fragment carries its own IP header. At the destination, the Network Layer reassembles the fragments using the identification number, fragment offset, and more-fragment flag to reconstruct the original packet.

4. Sliding Window Protocol

Given:

- Window size = 6 frames
- Total frames = 48

Calculation:

- Number of windows = $48 \div 6 = 8$ **windows**
- One frame lost per window
- Total retransmissions = **8 frames**

Explanation:

Flow control regulates the rate of data transmission between the sender and receiver. It prevents data loss, avoids receiver overload, and ensures efficient communication.

5. Session Layer Synchronization

Given:

- Video size = 600 MB
- Checkpoint every 60 MB

- Failure after 390 MB

Calculation:

- Number of checkpoints before failure = $390 \div 60 = 6$ **checkpoints**
- Last checkpoint = **360 MB**
- Data to retransmit = $390 - 360 = 30$ **MB**

Explanation:

The **Session Layer** inserts synchronization checkpoints that allow communication to resume from the latest checkpoint after a failure. This reduces retransmission time and improves efficiency.

6. Presentation Layer Compression and Encryption

Given:

- Original file size = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Calculation:

- Compressed file size = $900 \times 60\% = 540$ **MB**
- Encrypted file size = $540 \times 1.05 = 567$ **MB**
- Total reduction = $900 - 567 = 333$ **MB**
- Percentage reduction = $(333 \div 900) \times 100 = 37\%$

Explanation:

The **Presentation Layer** compresses data to reduce transmission time and bandwidth usage. It also encrypts data to protect confidentiality and ensure secure communication.

7. FTP Data Transfer

Given:

- File size = 3 GB = 3000 MB
- Request size = 3 MB
- Throughput = 120 Mbps

Calculation:

- Number of requests = $3000 \div 3 = 1000$ requests
- Total data = $3000 \times 3 = 9000$ MB
- Transmission time = $9000 \div 120 = 75$ seconds

Explanation:

The **Application Layer** provides services such as file transfer, authentication, and user communication. FTP ensures reliable file transfer between client and server.

8. End-to-End Delay

Given:

- Application Layer = 4 ms
- Transport Layer = 3 ms
- Network Layer = 5 ms
- Data Link Layer = 2 ms
- Physical Layer = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

Calculation:

$$\text{Total delay} = 4 + 3 + 5 + 2 + 1 + 18 + 12 = \mathbf{45 \text{ ms}}$$

Explanation:

Each OSI layer contributes to communication. The Application Layer provides services, the Transport Layer ensures reliable delivery, the Network Layer performs routing, the Data Link Layer handles framing and error detection, and the Physical Layer transmits bits. Propagation and transmission delays contribute to the overall end-to-end delay.

9. Frame Overhead

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead = 80 bytes/frame

Calculation:

- Useful data per frame = $1600 - 80 = 1520$ bytes
- Number of frames = $80,000,000 \div 1520 = 52,631.58 \approx 52,632$ frames
- Total overhead = $52,632 \times 80 = \mathbf{4,210,560 \text{ bytes } (\approx 4.21 \text{ MB})}$
- Transmission efficiency = $(1520 \div 1600) \times 100 = \mathbf{95\%}$

Explanation:

Protocol overhead reduces the amount of useful data transmitted in each frame. Although overhead consumes bandwidth, it is necessary for addressing, synchronization, and error detection.

10. Hospital Medical Record Transfer

Given:

- Original file size = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Data Link overhead = 60 bytes/frame

Calculation:

- Compressed file size = $240 \times 70\% = 168 \text{ MB}$
- Compressed size = **168,000,000 bytes**
- Number of segments = $168,000,000 \div 1400 = 120,000 \text{ segments}$
- Total protocol overhead = $120,000 \times 60 = 7,200,000 \text{ bytes (7.2 MB)}$
- Final data transmitted = $168 \text{ MB} + 7.2 \text{ MB} = 175.2 \text{ MB}$

Explanation:

The **Presentation Layer** compresses the data to reduce size. The **Transport Layer** divides it into segments and ensures reliable delivery. The **Network Layer** routes packets to the destination. The **Data Link Layer** adds frames and performs error detection. Finally, the **Physical Layer** transmits the complete data over the communication medium.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

and coherence

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand